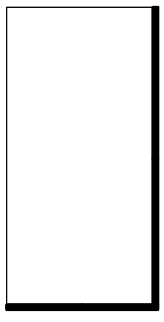
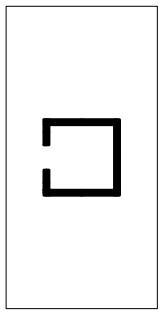
Task 1 Bracing

The subtasks a) to f) show different layouts of walls acting as bracing schemes of a plate. Tick the box if the bracing is working.

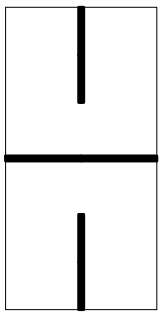
a) ☐



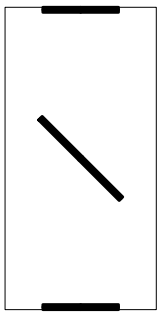
b) ☐



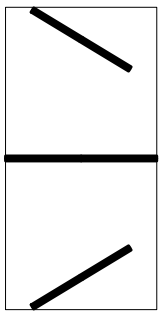
c) ☐



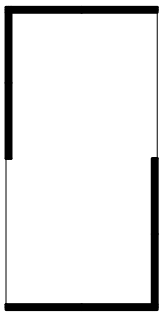
d) ☐



e) ☐



f) ☐



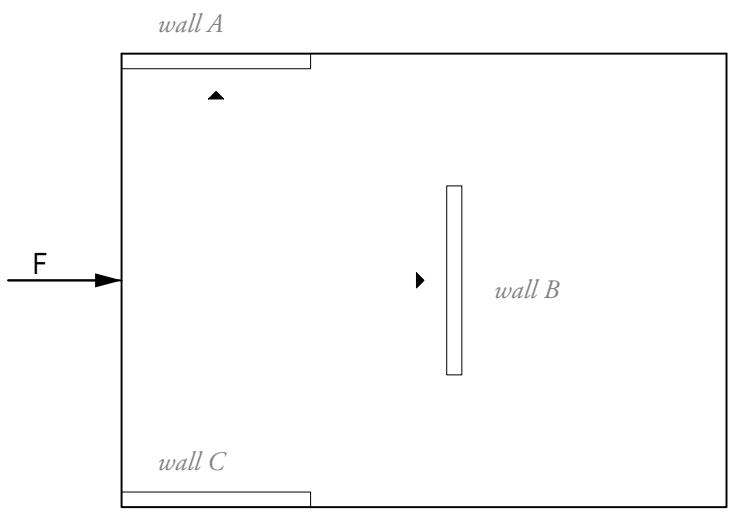
Task 2 Horizontal Forces (qualitatively)

Analyse the force flow within the ceiling due to an applied horizontal force. The walls are used for bracing.

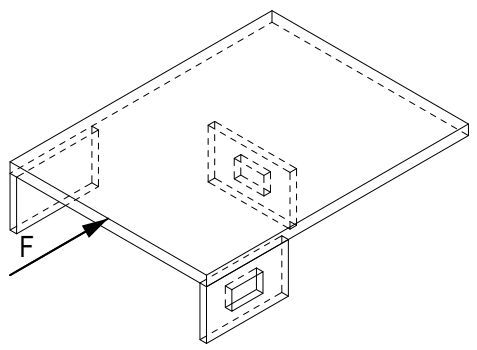
a) Find a qualitative internal force flow in the plate such that it can be redirected into the ground through the walls.

b) Then find a possible internal force flow in the walls A, B and C. First draw the applied horizontal force for each wall into the corresponding form diagram. Then, find the support forces. Indicate tension forces with red, compression forces with blue and reaction forces with green.

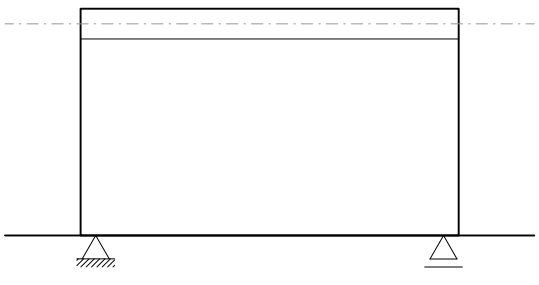
a)



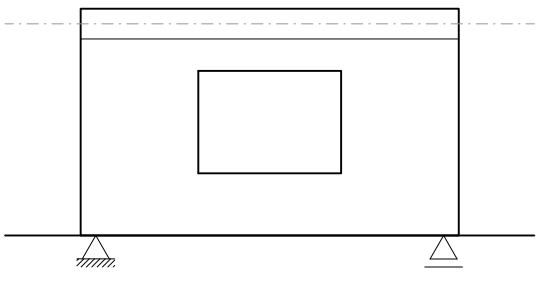
top view 1:200



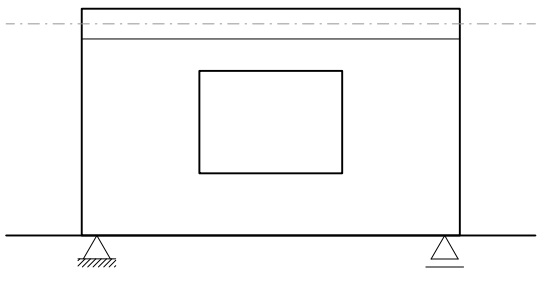
b)



form diagram wall A 1:100



form diagram wall B 1:100



form diagram wall C 1:100